

Oluwatimilehin (Timi) Emmanuel Owolabi

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Education

Covenant University

Bachelor of Engineering in Electrical & Electronics Engineering, GPA: 4.85/5.0

Ogun, Nigeria

Sep 2020 – Aug 2025

- **Thesis:** [Coordinated Control of Quadrotor Swarms: Classical, Adaptive, and Learning-Based Methods](#)

Technical Skills

Languages: Python, C++, SQL

ML & Robotics: PyTorch, TensorFlow, JAX, Hugging Face, MLflow, FastAPI, LangChain, ROS2, MuJoCo, CasADi

DevOps & Cloud: Docker, Kubernetes, AWS, GCP, Helm, Terraform, GitHub Actions, Argo CD, Grafana, OpenTelemetry

Professional Experience

Quidax Technologies

Lagos, Nigeria (Remote)

Platform & AI Engineer (Graduate Trainee)

Jan 2026 – Present

- Optimized AWS EKS autoscaling with Karpenter, increasing CPU utilization from 20% to 50% and memory utilization from 30% to 70%, reducing infrastructure costs by 8% while maintaining reliability.
- Modernized legacy manual deployment workflows by migrating Node, Python, Go, and Java services to reusable GitHub Actions and Argo CD pipelines, helping achieve 98% automated deployment/failover coverage across 1,099 jobs.
- Created and configured dedicated non-exchange Traefik ingress classes across development, staging, and production, separating non-exchange services from exchange resources and reducing traffic through the core exchange load balancer.
- Enabled mobile telemetry ingestion for the App team by developing and deploying authenticated OTLP proxies that forwarded traces, logs, and metrics to internal OTEL collectors and observability backends.
- Built an isolated sandbox environment for merchants to safely test integrations and workflows without impacting production.
- Instrumented the merchant platform with distributed tracing and logging using OpenTelemetry, enabling engineers to root-cause production incidents independently and reducing cross-team escalations.
- Led the engineering track of QuidaxOS, an AI agent that supports Engineering, Product, and Design teams from product discovery to deployment, reducing handoff friction and aligning execution with company objectives.

Trotta Technologies, Inc

San Francisco, California (Remote)

Founding Machine Learning Engineer

Jan 2026 – Feb 2026

- Founding engineer and ML team lead on a short-term contract, building the company's AI infrastructure on Google Cloud Vertex AI and shipping a production multi-agent defense system for phishing, deepfake audio, and synthetic video detection.
- Developed a self-improving AI threat detection system using LangSmith-hosted agents and anomaly detection neural networks on Google Cloud Run, reducing malicious email and phone call verification time from about 1 hour to under 20 seconds while enabling continuous self-supervised model retraining from production signals.

Scrella Technologies

Lagos, Nigeria

Lead Machine Learning Engineer

Jul 2024 – Dec 2024

- Led a three-person ML team and built the company's AI infrastructure on AWS SageMaker, architecting the full MLOps lifecycle from data collection through production deployment.
- Designed and deployed an AI-powered phone inspection pipeline using fine-tuned YOLOv11, CLAHE, spectral analysis, and Google PaliGemma to detect device imperfections from user videos, eliminating manual approval and reducing onboarding review time from 5 minutes to 18 seconds.

Peak Fiction

Belfast, United Kingdom (Remote)

Machine Learning Engineer

Apr 2024 – Jul 2024

- Utilized Latent Diffusion Models to create high-quality comic book scenes.
- Engineered an Image Generation workflow using ComfyUI, increasing productivity by 50%.

Schneider Electric

Lagos, Nigeria

Sustainability Intern

Aug 2023 – Oct 2023

- Conducted robotics workshops for young students on building a self-driving car.
- Partnered with NGOs to deliver STEM tutoring aligned with Schneider Electric's global initiative to empower 1 million youths.
- Reviewed NGO support budgets exceeding \$60,000 each, ensuring alignment with sustainability program objectives.

Research Experience

ML Collective

Lagos, Nigeria

Embodied AI Research Group Lead

Jun 2026 – Present

- Optimized large world-action models for efficient inference on resource-constrained edge devices.

- Implemented layer skipping on NVIDIA's Cosmos3-Edge-DROID model to reduce inference latency, developing a compact world-action model called SmolWAM.

WaySense — Dr. Daniel Omeiza, Oxford Robotics Institute

Oxford, United Kingdom (Remote)

Undergraduate Research Assistant

Mar 2024 – Sep 2024

- Analyzed the Lyft Level 5 dataset to estimate multi-agent vehicle positions and velocities, detect lane-change maneuvers, and improve explainability of autonomous driving decisions using ego- and agent-centric representations.
- Pioneered adaptation of the fast subset-scan algorithm on Graph Neural Networks to identify statistically significant anomalies in traffic datasets.

National Centre for Artificial Intelligence and Robotics (NCAIR)

Abuja, Nigeria

Research Intern

Dec 2025 – Present

- Developed AI-powered assistive glasses for visually impaired users, deploying Qwen3.5-0.8B on a Raspberry Pi via llama.cpp for on-device scene understanding and real-time environmental descriptions.

Google Developer Groups on Campus

Ogun, Nigeria

Robotics Research Team Lead

Sep 2024 – Aug 2025

- Led replication of Stanford's ALOHA bimanual robotic manipulation paper in simulation.
- Developed vision-based manipulation policies for box-picking and object interaction tasks using MuJoCo and JAX.
- Prototyped a low-cost 3D-printed robot arm for object sorting, matching WidowX 250 S performance at 2.5% of its cost.

Covenant University Centre for Research, Innovation and Discovery

Ogun, Nigeria

Research Student

Sep 2024 – Aug 2025

- Collaborated on multidisciplinary AI research projects including autonomous garbage collection systems and epidemiological prediction models for pandemic preparedness in Sub-Saharan Africa using machine learning approaches.
- Developed an intelligent research discovery system using natural language processing to automatically parse, summarize, and cluster academic literature from Scopus alerts, significantly improving research workflow efficiency.

Zummit Africa

Lagos, Nigeria

Machine Learning Research Intern

April 2023 – May 2024

- Developed a brain tumor segmentation application, achieving 95% accuracy using U-Net and medical imaging datasets.
- Developed a toxic comment classifier with imbalanced social media data; benchmarked BERT against BiLSTM.

Innovia Labs

Lagos, Nigeria

Research Intern

Mar 2024 – May 2024

- Contributed to the development of an electric VTOL drone for search and rescue applications by configuring the Pixhawk4 flight controller for autonomous flight.
- Developed a gesture-controlled home automation system using a cascaded landmark detection model and multilayer perceptron (MLP) classifier.

EPAIL Nigeria

Lagos, Nigeria

Research Intern

Aug 2023 – Oct 2023

- Developed and deployed a vision-based firearm detection system for security turret booths; collected and annotated a dataset of 10,000 firearm segmentation images.
- Developed a small-sized fixed-wing drone for surveillance operations.

Publications & Manuscripts

T. E. Somefun, **T. Owolabi**, and O. M. Longe, “[Practical Trade-offs in Neural Network Optimization: Brute Force Search and Gradient Descent](#)”, *Engineering Research Express*, 2025.

A. Awelewa, K. Ojo, T. Abimbola-Oladejo, **T. Owolabi**, “[Fuzzy-PID Controller for Liquid Level Control of Tank Systems](#)”, *NIPES-Journal of Science and Technology*, 2025.

T. E. Somefun et al. and **T. Owolabi**, “[Energy Optimization Algorithm for Reducing Energy Consumption in a Smart Home](#)”, *IEEE ICMEAS*, 2023.

O. Idowu et al. and **T. Owolabi**, “[Enhancing Radiological Imaging for Better Healthcare Outcomes Through High Performance Hybrid Approach](#)”, *ASRIC Journal on Engineering Sciences*, 2024.

O. Owolabi, C. Chukwuma, O. Omogboyega, and A. Awelewa, “[Coordinated Control of Quadrotor Swarms: Classical, Adaptive, and Learning-Based Methods](#)”, *Under Review, IJCAI*, 2026.

A. Awelewa, K. Ojo, E. Edmond, **O. Owolabi**, and I. Samuel, “[Smart Energy Metering and Monitoring System Using Internet of Things](#)”, *Under Review, IEEE SAUPEC*, 2026.

Selected Projects

SmolWAM | Python, LIBERO, RoboCasa

Ongoing

- Developing SmolWAM, a compact world-action model based on NVIDIA Cosmos3 Policy for fast edge device inference.

- Benchmarking manipulation performance and inference efficiency across LIBERO and RoboCasa.
- Autonomous Quadcopter** | C++ , ESP32, PlatformIO 2025
- Implemented an ESP32 flight-controller with modular drivers for sensors, receiver input, state estimation, PID control, motor actuation, battery monitoring, EEPROM, and telemetry.
 - Built cascaded attitude/rate PID control, 50 Hz 16-bit ESC PWM motor mixing, GPS/barometer/IMU/compass sensor fusion, PPM receiver decoding, and a ground-station UI.
- Coordinated Control of Multi Quadrotor Swarms** | Python, PyTorch, JAX, PyBullet 2025
- Modeled quadrotor dynamics in quaternion and Euler state-space form, then implemented PID, LQR with integral action, MRAC, PPO, and MPC controllers for position and trajectory tracking.
 - Developed minimum-snap trajectory planning and multi-agent formation controllers using leader-follower, consensus, behavioral rules, and potential-field collision avoidance in gym-pybullet-drones.
- Pick-and-Place Robot Arm** | Python, JAX, MuJoCo, Gymnasium, Stable-Baselines3 2025
- Built custom Gymnasium environments around a MuJoCo Franka Panda scene, exposing actuator control ranges and observations from qpos, qvel, cube pose, end-effector pose, and finger poses.
 - Trained staged PPO policies for reaching, grasping, and lifting with reward shaping for distance, grasp quality, contact-based grasp detection, cube height, drop penalties, randomized resets, and success-rate evaluation.

Awards and Honors

- Semifinalist**, Africa Deep Tech Challenge 2026
- First Class Honours**, Dept. of Electrical and Information Engineering, Covenant University 2025
- Finalist**, Pan-African Robotics Competition 2024
- Founder’s Award**, Best WASSCE Result, Wellspring College 2021
- Valedictorian**, Department of Science, Wellspring College 2020

Selected Talks

- AI: Our Current Reality and Future Trajectory** Nigerian Society of Engineers, August 2024
- Deep Learning: Neural Networks** APWEN, April 2024

Teaching, Leadership & Community Impact

- Engineering Innovation Empowerment Foundation (EIEF)** Abuja, Nigeria
- Co-Founder** Dec 2024 – Present
 - Designed and produced hands-on engineering kits to teach core engineering principles.
 - Organized and led technical training programs, mentoring students in practical engineering skills.
 - Launched The Classroom Project to provide tutoring, mentoring, and learning materials for low-resource public secondary schools.
- Association of Electrical and Information Engineering Students** Ogun, Nigeria
- President** Sep 2024 – Aug 2025
 - Led a 20-member executive team in coordinating projects and competitions for over 2,000 students.
 - Launched a student mentorship program pairing juniors with seniors for academic and career guidance.
- Association of Professional Women Engineers of Nigeria** Lagos, Nigeria
- Machine Learning Tutor** Aug 2023 – Oct 2023
 - Taught foundational machine learning concepts, including supervised learning, model evaluation, and neural networks, to over 50 female engineering students.
 - Developed hands-on Python exercises and guided students through implementing and evaluating machine learning models on real-world datasets.
- Project Cage Technologies** Lagos, Nigeria
- Robotics Instructor** Aug 2023 – Oct 2023
 - Instructed students in robotics fundamentals, covering electronics, sensors, actuators, embedded programming, and autonomous systems.
 - Mentored students through hands-on robotics projects, guiding system design, hardware integration, programming, and troubleshooting.
- Robotics and Artificial Intelligence Innovators Network** Ogun, Nigeria
- Robotics and AI Team Lead** Sep 2024 – Aug 2025
 - Led the development of a speech emotion recognition system with PyTorch Lightning, using a CNN-LSTM pipeline to classify audio samples.
 - Organized a fruit image classification competition featuring ResNet-18, MobileNet V2, and EfficientNet V2; evaluated participants’ models and demonstrated why EfficientNet V2s achieved superior performance.

Enactus

Member (National Champions)

Ogun, Nigeria
May 2021 – May 2023

- Designed a soil-nutrient sensor circuit for the “Cow Blood to Manure” sustainability project.
- Contributed to the Pet City initiative which uses recycled plastic bottles to make interlocking road tiles.
- Supported the Bargass project which involves converting plant biomass waste into reusable food packs.

Society Memberships

ML Collective, since May 2026

Black in AI (BAI), since Apr 2025

Black in Robotics, since Apr 2025

African Robotics Network (AfROB), since Aug 2025

International Society of Automation (ISA), since May 2024

Association of Professional Women Engineers of Nigeria (APWEN), since Apr 2024

Data Science Nigeria(DSN), since Jun 2022

Nigerian Society of Engineers (NSE), since Dec 2021